

# Umesh Kashyap

✉ [umeshk@iitbhilai.ac.in](mailto:umeshk@iitbhilai.ac.in)    📍 Bhilai, India    [in LinkedIn](#)    [GitHub](#)    [G Google Scholar](#)

## PROFILE SUMMARY

---

Ph.D. Scholar at the **Indian Institute of Technology (IIT) Bhilai** with 4+ years of research experience in **Trustworthy AI** and **Adversarial Machine Learning**. Specialized in developing robust defense mechanisms against adversarial attacks, perceptual privacy protection, and deepfake detection.

Proven track record of publishing in high-impact venues (IEEE Transactions, SPACE) and securing competitive research funding (TIH Fellowship, NFOBC). Seeking a Post-Doctoral position to leverage expertise in secure deep learning architectures and contribute to cutting-edge research in AI safety.

## EDUCATION

---

**Ph.D. in Computer Science (Ongoing)** 2022 - Present  
Indian Institute of Technology Bhilai, India  
*Thesis Focus: Adversarial Security Evaluation & Privacy-Preserving Deep Learning*

**M.Sc. Computer Science** 2018 - 2020  
Atal Bihari Vajpayee University, Bilaspur, India

**B.Sc. Computer Science** 2015 - 2018  
Bilaspur University, Bilaspur, India

## RESEARCH INTERESTS

---

- **Adversarial Machine Learning:** Attacks (Patch, Gradient-based) and Defense mechanisms.
- **Privacy-Preserving AI:** Perceptual encryption, secure inference, and data privacy.
- **Forensics:** Deepfake detection, attribution, and style-based anomaly detection.
- **Trustworthy AI:** Robustness evaluation and model security benchmarking.

## PUBLICATIONS

---

1. R. Kumar, **U. Kashyap**, S. S. Ali, “Gradient-Guided Adversarial Patch Attack for Deep Neural Networks.” **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 227–245, 2026.
2. **U. Kashyap**, S. K. Padhi, S. S. Ali, “Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods.” **IEEE Transactions on Artificial Intelligence**, 2025.
3. A. Vishwakarma, **U. Kashyap**, S. S. Ali, “Adversarial Malware Detection.” **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 277–286, 2024.
4. S. K. Padhi, H. Kumar, **U. Kashyap**, S. S. Ali, “De-Fake: Style-Based Anomaly Deepfake Detection.” preprint arXiv:2507.03334, 2025.
5. S. M. Shahid, S. K. Padhi, **U. Kashyap**, S. S. Ali, “Generalized Deepfake Attribution.” preprint arXiv:2406.18278, 2024.

## FELLOWSHIPS & AWARDS

---

- **TIH PhD Fellowship:** Awarded by the Technology Innovation Hub (TIH) for research on AI-driven GST fraud detection (Dec 2023 – May 2025).
- **National Eligibility Test (UGC-NET):** Qualified **6 times** (2019–2025), demonstrating consistent top-tier academic competency nationwide.
- **NFOBC Doctoral Fellowship:** Awarded by University Grants Commission (2023, 2024).
- **CG-SET Qualified:** Chhattisgarh State Eligibility Test (2019).

## RESEARCH & PROFESSIONAL EXPERIENCE

---

**PhD Research Scholar** Jan 2022 - Present  
*MIST Lab, IIT Bhilai*

- Led the development of a security benchmark for perceptual encryption, identifying critical vulnerabilities in existing state-of-the-art methods.
- Designed “De-Fake,” a novel style-based anomaly detection framework for identifying deepfakes, improving detection rates on unseen datasets.

- Conducted extensive adversarial evaluation on malware detection systems, proposing robust countermeasures against evasion attacks.

### **Project Scientist**

*Jan 2023 - Dec 2023*

*Indian Institute of Technology (IIT) Delhi*

- Collaborated on the evaluation of next-generation wireless communication systems using ML-driven optimization techniques.
- Validated performance metrics through rigorous experimental setups and produced technical documentation for grant compliance.

### **Guest Faculty (Computer Science)**

*2020 - 2022*

*Atal Bihari Vajpayee University*

- Taught core curriculum: Machine Learning, Data Structures, and Programming.
- Mentored undergraduate students on capstone projects focusing on applied ML.

## **TECHNICAL SKILLS**

---

|                              |                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------|
| <b>Languages</b>             | Python, C, C++, LaTeX, SQL                                                            |
| <b>ML Framework</b>          | PyTorch, TensorFlow, Keras, Scikit-learn, Numpy, Pandas, Seaborn, Matplotlib, Open-CV |
| <b>Tools &amp; Platforms</b> | Git, Docker, Linux, Google Colab, Jupyter, Hugging Face                               |
| <b>Core Competencies</b>     | Model Robustness, Adversarial Attack/Defense, Computer Vision, NLP                    |

## **LANGUAGES**

---

**Hindi:** Native    **English:** Professional